

MTH 211: THEORY OF STATISTICS

L-T-P-D-C (3-1-0-0-11)

First Course Handout

A. Objectives:

MTH211: Theory of Statistics is a UG level course designed to provide students with a solid introduction to the principles and methods of parametric statistical inference. The primary aim of the course is to build a strong conceptual and mathematical foundation that enables students to understand how statistical models are developed, analyzed, and applied to real-world data.

Through this course, students will acquire the essential background needed to successfully pursue higher-level and advanced courses in both theoretical and applied areas of Statistics and Data Science. In addition, the course emphasizes the development of critical thinking skills required to analyze datasets, draw valid conclusions, and make informed, data-driven decisions. By the end of the course, students will be well prepared to engage with more advanced topics and to undertake meaningful research in statistics and related fields.

B. Prerequisites:

MSO205A: Introduction to Probability Theory and **MTH207A: Matrix Algebra and Linear Estimation (Module II)** are pre-requisites for students of BS Statistics & Data Science program for whom it is a compulsory course. The course is also offered as OE/DE to students (instructor's prerogative) with good knowledge of basic Probability Theory, Descriptive Statistics and Distribution Theory.

C. Course Contents: (approximate number of lectures (L))

I. Introduction (1)

- Population and random sample
- Statistics and parameters

II. Revision of some topics from Probability Theory (4)

- Some special distributions
- Distributions of functions of random variables;
- Sampling distributions of sample means and sample variances and their functions – chi-square, Student's-t and F distributions
- Distribution of order statistics and their properties
- Important large sample results, like, convergence in probability and in distribution, weak law of large numbers, central limit theorem

III. Point Estimation (14)

- Exponential family of distributions
- Sufficiency and Minimal Sufficiency
- Completeness
- Unbiasedness and Uniformly Minimum Variance Unbiased Estimation
- Information Inequalities, Cramer-Rao Lower Bound
- Methods of Estimation – Maximum Likelihood and Method of Moments estimators and their properties

IV. Testing of Hypotheses (10)

- Null and Alternative Hypotheses, Simple and Composite hypotheses, Level of Significance, Size and Power of test
- Neyman-Pearson Lemma, Most Powerful and Uniformly Most Powerful Tests
- Likelihood Ratio Tests

V. Interval Estimation (2)

- Pivotal Statistics, Construction of Confidence Intervals

VI. Simple Linear Regression Model (3)

- Least Squares Estimation of model parameters
- Tests and confidence intervals of various functions of the parameters

VII. Bayesian Estimation (5)

- Concepts of loss, risk, prior and posterior distributions, loss functions
- Bayes estimator, Bayes risk
- Credible sets and intervals

D. Special Emphasis:

The course will place special emphasis on helping students grasp conceptually challenging topics and on demonstrating how these concepts can be effectively applied to real-world problems and practical scenarios.

E. Lectures, Tutorials - Schedule & Venue:

Lectures: W, Th, F
Schedule: 11:00 - 11:50 hrs
Venue: FB557

Tutorials: Tue
Schedule: 11:00 - 11:50 hrs
Venue: FB557

F. Office Hours:

There is no scheduled office hour.

Students should contact the instructor over email to schedule a time for discussion.

G. Evaluation Components & Policies: Exams, Quizzes, Assignments, Attendance, Participation, Honesty practices, Withdrawal (within the limits of DOAA Guidelines), etc.

I. Evaluation

Quiz 1 - 10%,

Mid-Sem Exam - 30%

Quiz 2 – 10%

End-Sem Exam - 50%

All quizzes will be pre-announced. Exam dates are decided centrally.

Lecture notes and problem sets will be regularly uploaded in Mookit.

II. Course Policies:

Proper measures will be adopted to fully control and curb cheating and other malpractices. In the event of any apprehension, such cases will be referred to the SSAC.

Students will be allowed to withdraw from the course as per guidelines set by the academic office.

As per a Senate ruling, attendance will be recorded through Acadly. Students are strongly advised to attend lectures and tutorials and participate and interact during tutorial sessions.

Students are expected to explore computational aspects and work out numerical problems on some topics, like various methods of estimation, in the Data Science Lab sessions.

H. Books & References:

1. Casella, G., Berger, R. L. (2021). Statistical Inference. Cengage Learning. (Textbook)
2. Rohatgi, V. K., Saleh, A. M. E. (2015). An Introduction to Probability and Statistics. John Wiley and Sons.
3. Hogg, R. V., McKean, J., Craig, A. T. (2005). Introduction to Mathematical Statistics. Pearson Education.
4. Ross, S.E. (2004). Probability and Statistics for Engineers and Scientists. Elsevier

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Session 2025-26, Semester II

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